Assignment

Module 3.5 Network Testing Practical

1. **Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**

```
Microsoft Windows [Version 10.0.26300.9711]
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C:\Users\shaik>ping 192.168.0.107

Pinging 192.168.0.107 with 32 bytes of data:
Reply from 192.168.0.107: bytes=32 time=2215ms TTL=64
Reply from 192.168.0.107: bytes=32 time=3ms TTL=64
Reply from 192.168.0.107: bytes=32 time=4ms TTL=64
Reply from 192.168.0.107: bytes=32 time=117ms TTL=64

Ping statistics for 192.168.0.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 2215ms, Average = 584ms
```

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2. **Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.**

```
C:\Users\shaik>tracert www.youtube.com

Tracing route to www.youtube.com [142.251.153.4]
over a maximum of 30 hops:

  1    10 ms     1 ms     2 ms  192.168.0.1
  2    29 ms     2 ms     1 ms  RTKGW.bbrouter [192.168.3.1]
  3    26 ms     6 ms     6 ms  100.113.0.0
  4     *        *        *     Request timed out.
  5    14 ms     7 ms     6 ms  100.100.107.239
  6     *        *        *     Request timed out.
  7    14 ms    14 ms    14 ms  172.31.22.76
  8    25 ms    13 ms    16 ms  112.133.203.98
  9    48 ms    14 ms    13 ms  142.251.153.4

Trace complete.

C:\Users\shaik>
```

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3. **Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.**

```
C:\Users\shaik>ping 192.168.0.107

Pinging 192.168.0.107 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.0.103: Destination host unreachable.
Reply from 192.168.0.103: Destination host unreachable.

Ping statistics for 192.168.0.107:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\shaik>ping 192.168.0.107

Pinging 192.168.0.107 with 32 bytes of data:
Reply from 192.168.0.103: Destination host unreachable.
Reply from 192.168.0.103: Destination host unreachable.
Reply from 192.168.0.103: Destination host unreachable.
Reply from 192.168.0.103: Destination host unreachable.

Ping statistics for 192.168.0.107:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

➢ After disconnecting the phone from the Wi-Fi network, the ping command was executed from the laptop using the phone's IP address. The request timed out, indicating that the laptop did not receive a response from the disconnected device. Thus, the ping test confirmed that the device was unreachable over the Wi-Fi network.

4. **Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.**

```
C:\Users\shaik>arp -a

Interface: 192.168.0.103 --- 0x3
  Internet Address      Physical Address      Type
  192.168.0.1           e8-65-d4-99-dd-88     dynamic
  192.168.0.105         a4-4e-31-b5-9f-20     dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0xd
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\shaik>
```

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Sr. No.	IP Address	MAC Address	Device
1	192.168.0.1	e8-65-d4-99-dd-88	Wi-Fi Router
2	192.168.0.105	a4-4e-31-b5-9f-20	My Laptop

5. **Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you**

**observed.

Hint: Include screenshots or command outputs if possible.**

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1. Connecting Devices to Wi-Fi

I connected my laptop and my friend's mobile phone to the same Wi-Fi network. I opened Command Prompt and ran the ipconfig command to check my laptop's network settings. My laptop's IP address was 192.168.0.105, and the router's IP address was 192.168.0.1.

After that, I used the ping command with the phone's IP address to check whether the two devices could communicate.

Result: The ping command helps determine whether another device can be reached through the network.

2. Checking the Route to YouTube

I entered the following command in Command Prompt:

traceroute www.youtube.com

This command shows the path that data follows from my laptop to YouTube. Each numbered step represents a hop through a router or another network device. It also displays the time taken for each step.

Result: The command helps identify the network path to YouTube. The number of hops varies depending on the internet connection and routing.

3. Checking Connectivity After Disconnecting a Device

I disconnected my friend's phone from the Wi-Fi network and tried to ping its IP address again using the following command:

ping [Phone IP Address]

If the laptop does not receive a response, the command may display "Request timed out." This can happen when the phone is disconnected or does not respond to ping requests.

Result: This test demonstrates how disconnecting a device can affect communication over a local network.

4. Finding Devices on the Network

I used the arp -a command to view the IP and MAC addresses stored in my laptop's ARP cache. The output showed the following information:

192.168.0.1 — Wi-Fi router.

192.168.0.105 — My laptop.

An IP address is used to identify a device on a network, while a MAC address identifies its network interface. I could check the router's connected-device list to find out which devices are connected.

Result: The ARP command helped display known network addresses, although it might not show every device connected to Wi-Fi.

5. Problems Encountered

During the practical, some network-related issues could occur. Ping requests might time out if a device does not respond. Traceroute might show timeouts when certain network devices block diagnostic requests. Also, the ARP output did not provide the names of all devices, making identification difficult.

These issues can be investigated by checking network connections, verifying IP addresses, and reviewing the router's device list.

This practical improved my understanding of basic networking and troubleshooting. I learned to use ipconfig to check network settings, ping to test connectivity, traceroute to examine the route to a destination, and arp -a to view known IP and MAC address mappings. These commands are useful for checking network communication and investigating common connectivity problems.